Report on

Early Detection of Autism from Facial Images

Submitted to the Department of Information Technology

Major Project, 7th Semester

Presented By:
Rohan Dutta (2021ITB009)
Rudra Roy (2021ITB010)
Dishita Barman (2021ITB058)

Under the supervision of
Dr. Shyamalendu Kandar
Assistant Professor
Department of Information Technology
IIEST, Shibpur

Department of Information Technology
Indian Institute of Engineering Science and Technology, Shibpur

December, 2024

Department Of Information Technology
Indian Institute of Engineering Science and Technology, Shibpur
Howrah - 711103

CERTIFICATE

This is to certify that the project entitled "**Early Detection of Autism from Facial Images**" submitted by **Rohan Dutta** (2021ITB009), **Rudra Roy** (2021ITB010) and **Dishita Barman** (2021ITB058) to Indian Institute of Engineering Science and Technology, Shibpur, towards partial fulfillment of the requirement for the award of degree of Bachelor of Technology in Information Technology is a record of bonafide work carried out by them under my supervision. This dissertation, in my opinion, is worthy of consideration for the purpose for which it has been submitted, and it fulfills the requirements of the regulations of this institute. The results incorporated in this dissertation are original and have not been submitted to any other university or institute for the award of any degree.

Dr. Shyamalendu Kandar
Assistant Professor
Department of Information Technology
Indian Institute of Engineering Science and Technology, Shibpur,
Howrah, India - 711103
December, 2024

ACKNOWLEDGEMENT

We would like to extend our heartfelt gratitude to **Dr. Shyamalendu Kandar**, our esteemed supervisor, for his invaluable guidance and steadfast support throughout the development of our report on “Early Detection of Autism and Classification using Transformer based Learning.” Dr. Kandar's expertise, encouragement, and constructive feedback have been instrumental in shaping the conceptualization, methodology, and execution of our project.

We are also thankful to the Department of Information Technology at the Indian Institute of Engineering Science and Technology Shibpur for providing the necessary resources and a conducive environment for research and learning. The collaborative spirit and academic excellence within the department have significantly contributed to the successful progression of our project.

This report reflects the culmination of our dedicated efforts and the mentorship provided by Dr. Kandar and the supportive academic community at our institution. We acknowledge the importance of this collaborative environment in fostering our understanding of machine learning algorithms for image classification, especially in the context of autism classification and its severity.

Rohan Dutta
(2021ITB009)

Rudra Roy
(2021ITB010)

Dishita Barman
(2021ITB058)

Abstract

Autism is a developmental handicap of children that gets worse as they age. An autistic child has problems with interaction and communication, as well as limited behaviour. If autistic children are diagnosed early, they can have a quality life by providing thorough care and therapy. However, in many developed countries, it is too late to diagnose children with autism. Besides, a trained medical expert is required to identify autism as there are no direct medical tests. Medical practitioners also take enough time to detect it because the children have to be monitored intensively.

In this project, machine and deep learning algorithms have been utilized for detecting autism in children from images that are not viable for ordinary people. We have employed four different algorithms that are Logistic Regression, Decision Trees, Random Forest (RF) and Convolutional Neural Network (CNN) for classifying Autism Spectrum Disorder (ASD) in children. Comparing classification performances among those algorithms, we have achieved the highest accuracy of 81.63% and a F1 Score of 84.02% on CNN, which outperformed the other conventional Machine Learning (ML) algorithms. Therefore, we proposed a prediction model based on CNN, which can be used for detecting ASD, especially for children.

Keywords: Autism Spectral Disorder (ASD), Early Detection of Autism, Classification Algorithms, Deep Learning, Convolution Neural Network, Self Attention.

List of Images

Image 1 Autistic Children Images 12

Image 2 Non Autistic Children Images 13

Image 3 Workflow 14

Image 4 Confusion Matrix for Logistic Regression 17

Image 5 Logistic Regression Training and Validation Metrics Over Epochs 17

Image 6 Decision Tree Training and Validation Metrics Over Epochs 21

Image 7 Confusion Matrix for Decision Tree 21

Image 8 Random Forest Training and Validation Metrics Over Epochs 24

Image 9 Confusion Matrix for Random Forest 24

Image 10 CNN Architecture Overview 27

Image 11 Custom CNN Architecture Training and Validation Metrics Over Epochs 31

Image 12 Confusion Matrix for CNN Model 33

List of Tables

Table 1 Performance Summary of Logistic Regression Classifier 18

Table 2 Performance Summary of Decision Tree Classifier 22

Table 3 Performance Summary of Random Forest Classifier 26

Table 4 Performance Summary of CNN Model Classifier 35

Table 5 Comparison Between Different model's performance 35

Chapter 1

Introduction

Autism is a developmental disorder that affects social interaction, communication, and learning, beginning in childhood and persisting into adulthood. It affects 1 in 54 children, with boys being four times more likely than girls to be diagnosed. Early diagnosis, ideally by age 2, enables timely intervention to improve behavioural and communication skills. However, diagnosing autism relies heavily on experienced physicians observing behaviour, which is time-consuming and inaccessible in many underdeveloped areas.

To address these challenges, researchers are developing machine learning (ML) models to detect autism using data from interviews and questionnaires. However, such methods are prone to errors due to children's nervousness or misunderstanding during these assessments.

This project introduces a simplified ML/DL-based approach for detecting Autism Spectrum Disorder (ASD) using facial images of 2,940 children aged 2–8 years. Four classification algorithms were tested, and the results showed that Convolutional Neural Networks (CNN) outperform other models in accurately classifying ASD. This method eliminates biases from interviews and provides a more efficient, reliable early-detection tool.

1.1 Autism and the Social Challenges It Faces

Autism Spectrum Disorder (ASD) is a complex neurodevelopmental condition characterized by difficulties in social interaction, communication, and repetitive behaviours. These challenges often manifest in various ways, significantly impacting the daily lives of individuals with autism. One of the most profound difficulties faced by children with autism is their struggle with understanding and expressing emotions, both their own and those of others. This impairment in emotional recognition and communication can lead to social isolation and difficulty in forming meaningful relationships, which are crucial aspects of human experience and development.

Children with autism frequently experience heightened levels of anxiety and frustration due to their inability to interpret social cues and respond appropriately in social situations. For example, they may find it challenging to recognize facial expressions, tone of voice, and body language, which are essential for effective. Communication and social interaction. This lack of understanding can result in misunderstandings and social rejection, further exacerbating feelings of loneliness and exclusion.

Moreover, the social challenges by children with autism extend beyond personal interactions. In educational settings, they may struggle to engage with peers and participate in group activities, which can hinder their academic and social development. The inability to interpret and express emotions can also affect their ability to collaborate with others. follow classroom routine. and respond to teachers' instructions, these difficulties can lead to behavioural issues, academic underperformance, and increased stress for both the child and their educators.

Families of children with autism also face significant social challenges. Parents and caregivers often experience high levels of stress and anxiety they navigate the complexities of raising a child with autism. They may encounter difficulties in finding appropriate educational and therapeutic resources, as well as in advocating for their child's needs within the

community and school systems. Additionally, social stigma and lack of awareness about autism can lead to misunderstandings and judgment from others, further isolating families and limiting their support.

Despite these challenges, there is growing recognition of the need for inclusive and supportive environments for individuals with autism. Advances in technology and research have opened new avenues for addressing the social challenges faced by children with autism. Interventions such as social skills training, behavioural therapy, and assistive technologies are being developed to help children with autism improve their social interactions and emotional understanding. By fostering a more inclusive and empathetic society, we can help children with autism overcome social barriers and lead more fulfilling lives.

In this context, our project aims to contribute to efforts by developing an Early Detection of Autism specifically designed for children with autism. By leveraging deep learning techniques, we seek to create a tool that can aid in the recognition and interpretation of facial emotions, thereby enhancing communication and social interaction skills in children with autism, this system has the potential to provide significant support in educational and therapeutic settings, ultimately helping children with autism navigate their social world with greater ease and confidence.

1.2 Autism Detection in Early Stages

Autism Spectrum Disorder (ASD) is a neurobehavioral disorder that begins at childhood and exists this whole life. Autism Detection in early stages is a technological advancement aimed at identifying autism spectrum disorder (ASD) during critical developmental phases. By utilizing computational models, researchers and practitioners can detect behavioural and neurological patterns indicative of ASD, enabling timely intervention and improved outcomes. This interdisciplinary field combines expertise from computer science, psychology, and neuroscience to develop systems that analyse diagnostic data and uncover subtle indicators of autism.

Our work introduces a custom architecture that combines convolutional neural networks (CNNs) for feature extraction with a self-attention mechanism to refine focus on critical diagnostic patterns. This integration enhances the model’s ability to prioritize relevant aspects of the data, ensuring higher accuracy and robustness.

The model processes key markers such as atypical gaze behaviour or developmental anomalies that are difficult to identify using traditional diagnostic tools. Automated feature extraction allows the system to analyse nuanced signals, bridging gaps in conventional methods and improving scalability in underserved regions. This advancement enables clinicians to design personalized treatment plans and empowers healthcare systems with tools for early and efficient ASD detection.

By incorporating self-attention mechanisms within a CNN-based framework, this work represents a significant step forward in medical diagnostics. It demonstrates how artificial intelligence can transform healthcare by offering reliable, scalable, and inclusive solutions. As these systems continue to evolve, they are poised to play a crucial role in bridging the gap between traditional clinical practices and emerging technologies, fostering a more inclusive and effective approach to developmental health.

1.3 Importance of Early Autism Detection in Children with ASD

Early detection of Autism Spectrum Disorder (ASD) is critical as it enables timely interventions during a child's formative years, when neuroplasticity is at its peak. Leveraging machine learning (ML) models, particularly deep learning architectures like convolutional neural networks (CNNs) integrated with self-attention mechanisms, provides a robust framework for identifying subtle diagnostic markers that are often challenging to discern through conventional methods. These models analyse diagnostic features such as atypical eye-gaze patterns, facial expressions, and developmental anomalies, which serve as early indicators of ASD. The ability of self-attention layers to focus on the most relevant aspects of input data further enhances the precision and reliability of the diagnostic process, reducing false positives and increasing confidence in early-stage diagnoses.

Integrating such models into therapeutic interventions and educational training programs significantly benefits children with ASD by enabling personalized, data-driven support. Early diagnosis allows for tailored intervention plans, including behavioural therapies and educational strategies, which are designed to address the specific developmental needs of the child. For instance, speech therapy or occupational therapy can be introduced earlier, targeting the precise areas where delays or deficits are detected by the ML system. The adaptive capacity of these models ensures continuous refinement based on the child's progress, offering dynamic feedback loops for both clinicians and caregivers.

Incorporating machine learning-powered autism detection into educational training programs can further enhance social and cognitive skills, equipping children with ASD to better integrate into mainstream environments. Predictive analytics derived from these models help educators identify the most effective teaching strategies and adjust them in real time, promoting skill acquisition and reducing the learning gap. Over time, these interventions contribute to improved communication, emotional regulation, and social interaction, helping children with ASD lead more fulfilling and independent lives.

The integration of machine learning into ASD detection and therapeutic systems exemplifies how technology can bridge gaps in traditional healthcare and education, ensuring more inclusive and effective care. As research advances, these systems are expected to become integral to multidisciplinary approaches, empowering children with ASD to achieve their full potential and enhancing their quality of life.

1.4 Approaches for Early Detection of Autism in Children

Early detection of Autism Spectrum Disorder (ASD) in children is a critical yet complex challenge that requires advanced technological approaches. Machine Learning (ML) and Deep Learning (DL) techniques have become pivotal in developing systems capable of identifying early indicators of ASD with precision. By leveraging large datasets annotated with autism-specific features, these models analyze patterns such as atypical gaze behavior, developmental anomalies, and other diagnostic markers to make reliable predictions. The integration of convolutional neural networks (CNNs) with self-attention mechanisms enhances the model's ability to prioritize relevant features, improving accuracy and robustness in detecting subtle signs of ASD. These advancements enable early intervention, which is vital for implementing tailored therapeutic and educational strategies that can significantly enhance the developmental outcomes and social integration of children with ASD.

Chapter 2

Literature Review

1. **A Convolutional Neural Network Model for Early-Stage Detection of Autism Spectrum Disorder [1]**

   Study employs Convolutional Neural Networks (CNN) to develop an AI-based framework for detecting Autism Spectrum Disorder (ASD) in children through facial image analysis. The approach aims to provide a non-invasive and efficient diagnostic tool to support medical practitioners, addressing the limitations and delays inherent in traditional behaviour-based diagnostic methods.

2. **Detection of autistic individuals using facial images and deep learning [2]**

   Introducing a novel approach, the study leverages deep learning to detect autism using facial images. It applies pre-trained models with transfer learning, incorporating advanced techniques like depth wise separable convolutions for computational efficiency. A rigorous data cleaning process using MD5 and perceptual hashing ensures accuracy by removing duplicate images. The method highlights the potential of facial feature analysis as an accessible and cost-effective alternative to traditional diagnostic methods.

3. **Hybrid Techniques of Facial Feature Image Analysis for Early Detection of Autism Spectrum Disorder Based on Combined CNN Features [3]**

   The study explores hybrid techniques for autism detection using facial images. It integrates deep learning models (VGG16, ResNet101, MobileNet) with machine learning classifiers (XGBoost, Random Forest). Key techniques include t-SNE for dimensionality reduction, feature fusion across models, and pre-processing with Gaussian filters and normalization to enhance image quality. This combination aims to extract and classify intricate facial features effectively.

4. **Early Detection of Autism in Children Using Transfer Learning [4]**

   The study employs transfer learning with a modified AlexNet architecture for early autism detection using facial features. Key techniques include replacing the final layers of AlexNet to adapt the model for binary classification and using pre-trained layers to extract general and class-specific features. The approach emphasizes minimal preprocessing and utilizes convolutional, pooling, and fully connected layers to classify autism-related traits effectively.

5. **Deep Learning Algorithms to Identify Autism Spectrum Disorder in Children-Based Facial Landmarks [5]**

   The study discusses the use of deep learning and machine learning models to detect autism spectrum disorder (ASD) in children based on facial landmarks. It employs MobileNetV2, a lightweight convolutional neural network (CNN), and a hybrid VGG-16 model combined with machine learning classifiers (e.g., logistic regression, SVM, random forest). The approach uses transfer learning to enhance feature extraction and optimize hyperparameters for better performance.

6. **Control learning rate for autism facial detection via deep transfer learning [6]**

   The research paper uses the DenseNet-121 model for autism detection based on facial images, employing transfer learning and a novel Control Sub gradient Algorithm (CSA) to dynamically adjust the learning rate for improved convergence and performance. L1 regularization further enhances generalization, outperforming traditional optimizers like SGD and Adam in ASD classification.

7. **Identification of Autism in Children Using Static Facial Features and Deep Neural Networks [7]**

   The paper investigates autism detection in children using facial features and pre-trained deep learning models as feature extractors, including MobileNet, Xception, and EfficientNet variants. A deep neural network (DNN) serves as the classifier. It employs transfer learning to leverage pre-trained models for effective feature extraction and optimized hyperparameters for accurate ASD classification. The study emphasizes the potential of static facial features as biomarkers for early autism screening.

8. **Empirical Study of Autism Spectrum Disorder Diagnosis Using Facial Images by Improved Transfer Learning Approach [8]**

   The paper explores the diagnosis of Autism Spectrum Disorder (ASD) using facial image analysis through deep learning. It employs transfer learning with several pre-trained convolutional neural networks (CNNs), including Xception, VGG19, ResNet50V2, MobileNetV2, and EfficientNetB0. The study modifies the Xception model, tunes hyperparameters, and optimizes settings for improved performance. It demonstrates the effectiveness of transfer learning in extracting discriminative features for ASD detection.

9. **Application of Machine Learning Techniques to Detect the Children with Autism Spectrum Disorder [9]**

   The paper investigates the application of machine learning to detect Autism Spectrum Disorder (ASD) in children by integrating physiological data (EEG) and behavioral data (eye fixation and facial expressions). It employs feature extraction techniques like k-means clustering for eye fixation and convolutional neural networks (CNNs) with soft labeling for facial expressions. A hybrid data fusion method based on a weighted Naive Bayes algorithm combines the modalities for improved classification performance, demonstrating the complementary strengths of the data types.

10. **Improved Transfer-Learning-Based Facial Recognition Framework to Detect Autistic Children at an Early Stage [10]**

    The paper proposes an improved transfer-learning framework using MobileNet-V1 for early autism detection through facial recognition. It integrates additional layers to enhance performance and employs k-means clustering to identify autism sub-types. Various machine learning and deep learning classifiers are compared, demonstrating the efficacy of the enhanced model for precise classification and clustering.

11. **Early Detection of Autism by Extracting Features: A Case Study in Bangladesh [11]**

    The paper explores early autism detection in Bangladesh using machine learning. It employs tree-based classification methods like J48, Random Forest, Logistic Model Tree (LMT), Reduced Error Pruned Tree (REPTree), and Decision Stump (DS). J48 was identified as the most effective classifier, and rules were extracted from the J48 decision tree to identify significant features and patterns associated with autism. The study focuses

on rule extraction, feature engineering, and regional analysis to improve early autism detection.

12. A Machine Learning Approach to Predict Autism Spectrum Disorder [12]

The paper introduces a prediction model for autism traits using a hybrid machine learning approach combining Random Forest-CART and Random Forest-ID3 classifiers. It leverages decision trees for feature selection and integrates the predictive model into a mobile application for user-friendly autism screening across different age groups. The model employs supervised learning techniques to optimize performance in identifying autism traits.

13.Identification of autism spectrum disorder using deep learning and the ABIDE dataset [13]

The paper utilizes a deep learning approach, specifically denoising autoencoders and a multilayer perceptron (MLP), to classify autism spectrum disorder using functional connectivity data from rs-fMRI. The method involves unsupervised pre-training to reduce data dimensionality and supervised fine-tuning for classification. This approach identifies neural connectivity patterns associated with ASD while accommodating multi-site data variability.

14. Automatic Identification of Down Syndrome Using Facial Images with Deep Convolutional Neural Network [14]

This paper presents a method for identifying Down syndrome using 2D facial images processed by a Deep Convolutional Neural Network (DCNN). The approach involves image preprocessing (enhancement, facial detection, cropping, resizing), transfer learning from general facial recognition to Down syndrome identification, and binary classification using a DCNN model. The DCNN architecture includes convolutional layers, inception blocks, and fully connected layers, optimized with transfer learning and regularization to prevent overfitting. This fully automated technique emphasizes subtle facial differences for efficient Down syndrome detection.

15. Automatic Detection of ADHD and ASD from Expressive Behaviour in RGBD Data [15]

This paper presents a novel automated approach for diagnosing ADHD and ASD using video analysis of expressive behaviour captured with RGB-D (color+depth) data. It employs a Dynamic Deep Learning-based system to analyse facial expressions, head motion, and pose. Feature extraction involves metrics like facial action units, Kinect-based animation units, head movement, and response times. Support Vector Machines (SVM) with feature selection are used for classification, aiming to predict these conditions from behavioural cues efficiently and objectively.

16. A Feasibility Study of Autism Behavioural Markers in Spontaneous Facial, Visual, and Hand Movement Response Data [16]

This paper proposes a novel approach for identifying behavioural markers of Autism Spectrum Disorder (ASD) using multimodal non-intrusive sensing techniques. It combines computer vision-based facial expression analysis (via Facial Action Coding System), eye-tracking, and hand movement data during interactive tasks. The system leverages Support Vector Machines (SVMs) with feature selection to classify facial action units and analyse correlations between eye gaze and hand motion to detect coordination deficits, aiding early ASD diagnosis and intervention planning.

CHAPTER 3

EARLY DETECTION OF AUTISM USING MACHINE LEARNING

In the first part of the project, we employed various basic machine learning classifiers to detect autism in children. This section details the methodologies, datasets used, preprocessing, results, and comparisons.

Overview of Basic Classifiers Used

1. **Logistic Regression**: A linear model used for classification tasks; it predicts probabilities using a sigmoid function and works well for simple, linearly separable data. However, it struggles with complex or non-linear patterns.

2. **Decision Tree**: A tree-based model that splits data into decision nodes based on feature thresholds, making it intuitive and capable of capturing non-linear relationships. However, it is prone to overfitting, especially with deep trees.

3. **Random Forest**: An ensemble method that builds multiple decision trees and combines their outputs to improve accuracy and reduce overfitting. It handles large datasets well but can be computationally intensive and less interpretable.

Overview of the Dataset used

In this project, we have used a publicly available dataset named Autistic children dataset collected from Kaggle repository. This dataset consists of **2940** face images of children where autistic and non-autistic classes are evenly distributed. In this dataset, the majority of the images are from **2 to 8** years children. The male and female ratio in the autistic class is **3:1**, and the non-autistic class is **1:1**.

Images of Autistic Children:

Images of Non-Autistic Children:

Overview of the Preprocessing Steps

In this study, the preprocessing of input images was a critical step to standardize and prepare the dataset for analysis and model training. The process involved three primary stages: resizing, grayscale conversion, and array transformation.

Resizing:

The dataset consisted of images with varying dimensions, which could introduce inconsistencies during model training. To ensure uniformity, all images were resized to a fixed dimension of 128x128 pixels using the resize function provided by Python's OpenCV library. This step not only standardized the input dimensions but also balanced the computational efficiency and retention of critical features within the images. The uniform size facilitates batch processing during training and ensures compatibility with the input layer of the deep learning models.

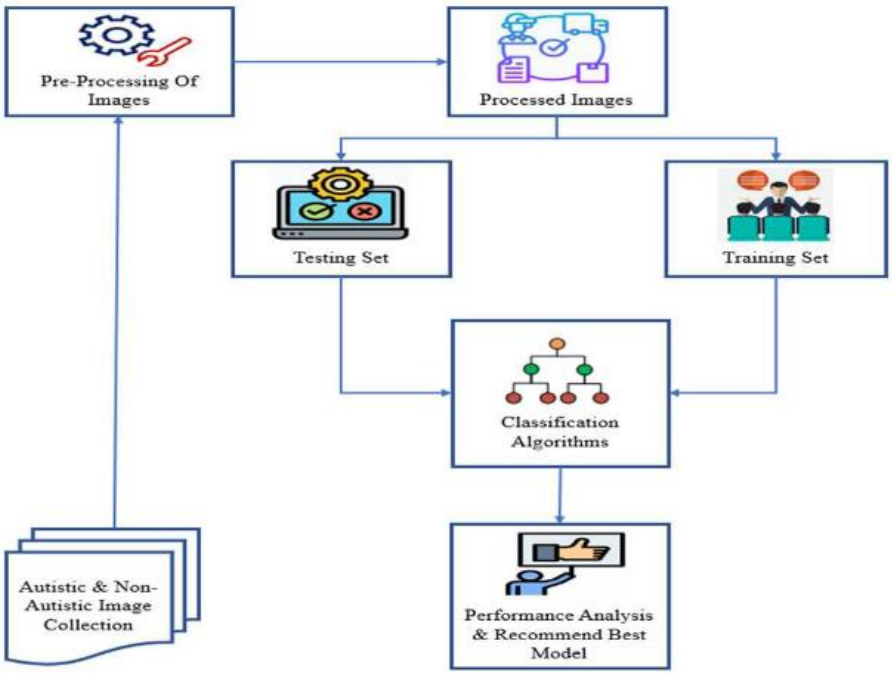
Grayscale Conversion:

Following resizing, the images were converted from the BGR (Blue-Green-Red) color space to grayscale using OpenCV's cv2.cvtColor function. Grayscale conversion simplifies the dataset by reducing the complexity of the input data from three color channels to a single intensity channel. This transformation minimizes computational overhead while preserving the essential structural and textural information of the images, which are often sufficient for tasks such as classification and feature extraction. By removing color information, the models focus solely on the intensity gradients and spatial patterns.

Array Transformation

The final step in the preprocessing pipeline involved transforming the processed images into numerical arrays, enabling their compatibility with machine learning and deep learning frameworks. Using NumPy, each image was represented as a two-dimensional array (for grayscale images) containing pixel intensity values normalized within a specific range, such as 0 to 255 or 0 to 1. This representation allows for seamless integration into tensor-based operations during model training and ensures that the input data adheres to the expected format of the employed algorithms.

Methodology Workflow



3. Machine Learning Models

3.1 Logistic Regression for Autism Classification:

3.1.1 Definition:

Logistic Regression is a supervised machine learning algorithm used for binary classification tasks. It models the probability of a binary outcome using a logistic function, predicting class labels based on input features. Logistic Regression is simple, interpretable, and effective for linearly separable data.

This approach suits tasks involving high-dimensional feature vectors like the Histogram of Oriented Gradients (HOG) features, ensuring efficient classification of facial images as either autistic or non-autistic.

3.1.2 Feature Extraction:

Histogram of Oriented Gradients (HOG):

- **Purpose:** Captures structural and edge information from images, emphasizing edges and contours.
- **Process:**

1. ***Divide the Image into Small Cells***

- **Description**: Partition the image into non-overlapping cells of size $\boldsymbol{c} \times \boldsymbol{c}$ pixels.
- **Mathematics**:
  - If the image dimensions are $W \times H$, the number of cells is:

$$\textbf{Number of Cells} = \frac{\mathbf{W}}{\mathbf{c}} \times \frac{\mathbf{H}}{\mathbf{c}}$$

2. ***Compute the Intensity Gradient for Each Pixel in the Cell***

- **Description**: Calculate the gradient magnitude and orientation for every pixel in the cell.

- **Mathematics**:
  - Gradient in x and y-directions:

$$G_x(x, y) = I(x + 1, y) - I(x - 1, y)$$

$$G_y(x, y) = I(x, y - 1) - I(x, y + 1)$$

  - Gradient Magnitude:

$$M(x, y) = \sqrt{G_x(x, y)^2 + G_y(x, y)^2}$$

  - Gradient Orientation:

$$\theta(x, y) = \tan^{-1}(\frac{G_y(x, y)}{G_x(x, y)})$$

3. ***Quantize Gradient Directions into Bins to Create Histograms***

- **Description**: Orientations are divided into b bins, and each pixel contributes to the histogram based on its gradient magnitude.
- **Mathematics**:
  - Binning range:
    Orientation bins: [0º,180º) or [0º,360º) (depending on symmetry)
  - Contribution to the histogram for pixel (x,y):

$$\text{Bin}(\theta(x, y)) \leftarrow \text{Bin}(\theta(x, y)) + M(x, y)$$

4. ***Normalize Histograms Within Neighboring Cells to Ensure Robustness***

- **Description**: Normalize histograms in overlapping blocks to make the descriptor invariant to lighting and contrast changes.
- **Mathematics**:

  - Concatenate histograms from $n \times n$ cells into a block vector v.
  - Apply **L2-Norm** normalization:

v^\prime=vv22+ϵ25. **Concatenate Block Histograms into a Feature Vector**

- **Description**: Flatten and concatenate all block histograms to create a single feature vector representing the image.

- **Mathematics**:
  - Feature vector dimension:

$$\text{Dimension} = (\frac{W}{c} - n + 1) \times (\frac{H}{c} - n + 1) \times n^2 \times b$$

    - n: Number of cells in a block.
    - b: Number of orientation bins.

Application in Autism Research: HOG is ideal for analyzing subtle facial features and expressions, focusing on structural patterns like eyes, nose, and mouth. This feature extraction method is robust to lighting and noise, making it suitable for analyzing variations associated with autism.

3.1.3 Training Methodology:

- ***Model Selection:***

  Logistic Regression was chosen for its interpretability and suitability for binary classification tasks. The model uses a sigmoid function to calculate probabilities and predicts labels based on a decision threshold.

- ***Hyperparameter Tuning:***

  Using **GridSearchCV**, key parameters were optimized:

- **C (Regularization Strength):** Prevents overfitting; tested values include 0.1, 1, and 10.
- **Penalty:** Tested l1 (LASSO, $\text{penelty Term} = \lambda \sum_{i=1}^{n} |\beta_i|$) and l2 (Ridge, $\text{penelty Term} = \lambda \sum_{i=1}^{n} {\beta_i}^2$) regularizations (where λ = hyperparameter and β_i = Model Coefficients.)

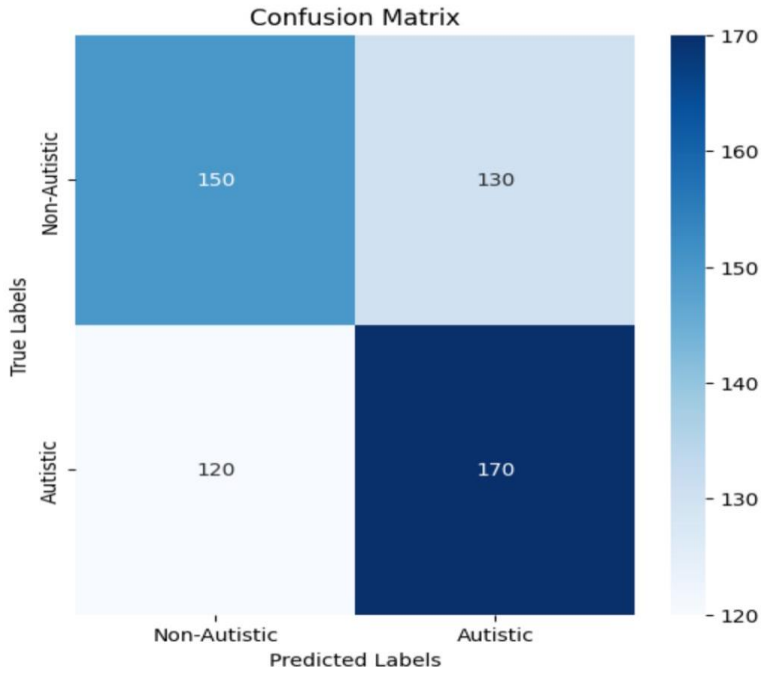
- **Solver: liblinear** for handling small datasets and sparse features.
- Cross-validation (CV) ensured model generalization by splitting the data into training and validation sets.

3.1.4 Performance Assessment

Evaluation Metrics

The performance of the model was evaluated using a variety of metrics to ensure a comprehensive understanding of its effectiveness:

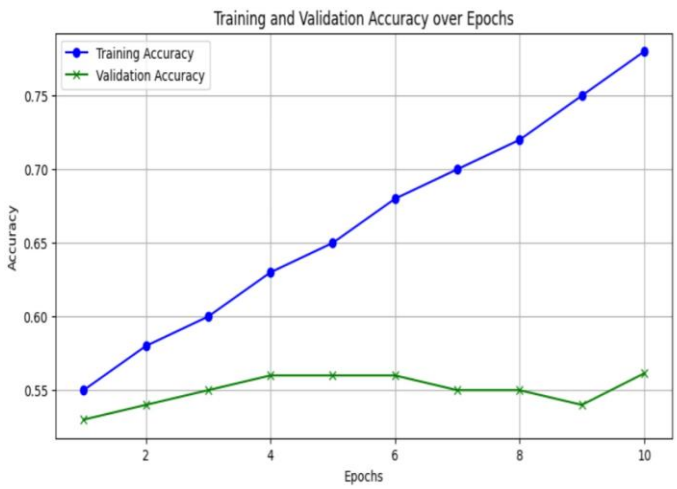
Confusion Matrix

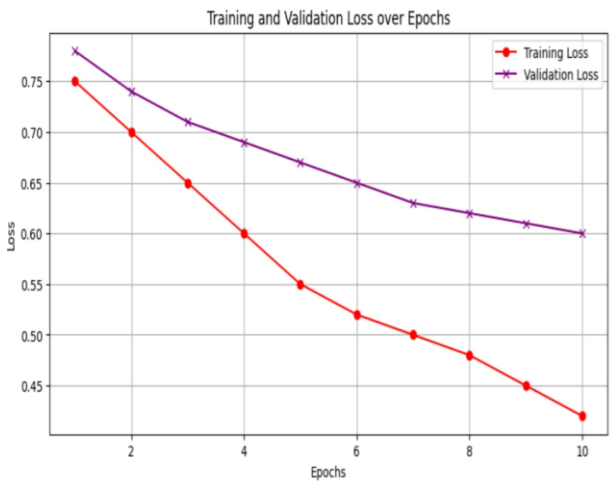


Result Over Validation Dataset

Metric	Value
Accuracy	56.14%
Precision	57.0%
Recall	59.0%
F1 Score	58.0%

- **Validation Curves**
  a. **Training vs. Validation Accuracy**: Plots showed consistent performance across grid search iterations, indicating a well-optimized model.
  b. **Loss Curves (1 - Accuracy)**: The curves revealed minimal overfitting, as the validation loss closely tracked the training loss.





3.1.5 Summary:

- **Strengths:** HOG features combined with Logistic Regression provided a moderately effective classification framework, leveraging edge information to distinguish between autistic and non-autistic images.
- **Challenges:** The model exhibited errors in both false positives and false negatives, suggesting room for improvement.
- **Next Steps:**
    - Explore advanced classifiers like Random Forests or Neural Networks.
    - Incorporate complementary features, e.g., color, texture, or spatial relationships, to enhance accuracy.

3.2 Decision tree:

3.2.1 Definition:

A **decision tree** is a supervised machine learning algorithm used for classification and regression tasks. It represents decisions in a tree-like structure, where each internal node splits data based on a feature, branches represent outcomes, and leaf nodes give the final prediction (class or value). The tree recursively partitions data into smaller subsets, aiming to make each subset more homogeneous regarding the target variable.

Decision trees are widely used in classification tasks because they are intuitive, easy to interpret, and can handle both numerical and categorical data without requiring extensive preprocessing. They work well for capturing nonlinear patterns and interactions between features, and they are robust to data with missing values. Their ability to prioritize important features and efficiently classify data makes them versatile tools in various machine learning applications.

3.2.2 Feature Extraction:

Histogram of Oriented Gradients (HOG) is a feature extraction technique used in computer vision to capture the structural and edge information in an image. It represents an image by analyzing the gradient directions (orientations) of intensity changes, which highlight edges, corners, and textures. HOG is particularly effective for detecting patterns like shapes or contours, making it suitable for tasks involving object detection and facial analysis.

HOG divides an image into small cells, computes the intensity gradient for each pixel, and quantizes the gradient directions into bins to create histograms for each cell. These histograms capture the distribution of edges and orientations within the cell. To ensure robustness to lighting and contrast changes, neighboring cells are grouped into blocks, and the histograms are normalized. Finally, all block histograms are concatenated into a single feature vector that represents the image.

In autism research, analyzing facial expressions and features is crucial for understanding differences in social and emotional cues. HOG is particularly useful because it focuses on edge and structural information, which are key to identifying facial characteristics like the eyes, nose, and mouth. Its robustness to lighting and noise ensures consistent feature extraction across diverse images, making it a reliable tool for studying subtle variations in facial features often associated with autism.

3.2.3 Training Methodology:

The goal of the training process was to create a Decision Tree Classifier capable of generalizing patterns in facial features. The model's interpretability and ability to handle both numerical and categorical data made it a suitable choice for this task.

- ***Model Initialization***

A Decision Tree Classifier was selected due to its simplicity, interpretability, and ability to capture complex, non-linear patterns in data. The tree works by recursively partitioning the dataset into smaller, more homogeneous subsets based on feature thresholds, eventually forming a tree structure where leaf nodes represent the predicted classes. This method ensures the model can efficiently identify patterns in facial features.

- ***Hyperparameter Tuning***

To optimize the classifier, **GridSearchCV** was used. This technique systematically evaluates combinations of hyperparameters to identify the best-performing setup. The key parameters considered were:

- **Criterion**:
- **Measures the quality of splits.**
  - ***Gini index*:** Focuses on reducing impurity quickly, with computational efficiency.

$$\mathrm{Gini(D)} = 1 - \sum_{i=1}^{c} p_i^2$$

where,
D : Dataset being evaluated
C : Number of Classes
p_i : Proportion of Instances in class i, calculated as
$p_i = \frac{count\ of\ class\ i}{Total\ Instances}$
for a split,

$$Gini_{split}(D) = \sum_{j=1}^{k} \frac{|D_j|}{|D|} Gini(D_j)$$

where,
$|D_j|$: number of instances in subset D_j.
|D| : total number of instances in D.

- ***Entropy and Information Gain*:** Uses information gain to measure the uncertainty reduction at each split.

$$\text{Entropy}(D) = -\sum_{i=1}^{C} p_i \log_2(p_i)$$

where,

p_i : proportion of instances in class i,

$$\boldsymbol{p_i} = \frac{\textit{count of class i}}{\textit{Total Instances}}$$

$$\text{Information Gain}\ (D, \text{split}) = \text{Entropy}(D) - \sum_{j=1}^{k} \frac{|D_j|}{|D|} \text{Entropy}(D_j)$$

where,

$|Dj|/|D|$: weight of subset D_j relative to the original dataset.

- **Max Depth**: Limits the depth of the tree to control its complexity and avoid overfitting. Values of 10 and 20 were tested to find a balance between capturing patterns and managing noise.
- **Min Samples Split**: Specifies the minimum number of samples required at a node for it to split further. Testing values of 2 and 5 ensured the tree would avoid overfitting by requiring a minimum sample size for splits.

The model was evaluated using **3-fold cross-validation**, which splits the data into three parts, using two for training and one for validation in each iteration. This ensured the model's performance was consistent across different data splits.

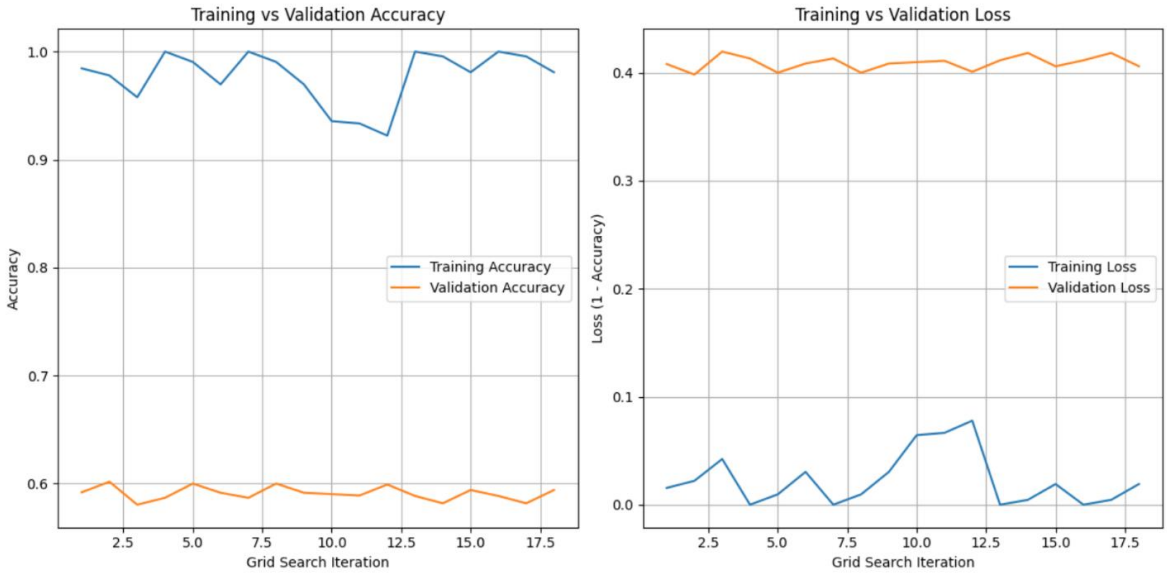
- ***Training with Optimal Parameters***

The best hyperparameters identified through **GridSearchCV** were:

- **Criterion**: **gini** (chosen for faster computation and effective performance on moderately sized datasets).
- **Max Depth**: 10 (enabled the model to capture intricate patterns in facial features without overfitting to noise).
- **Min Samples Split**: 8 (prevented the tree from splitting too frequently, ensuring better generalization).

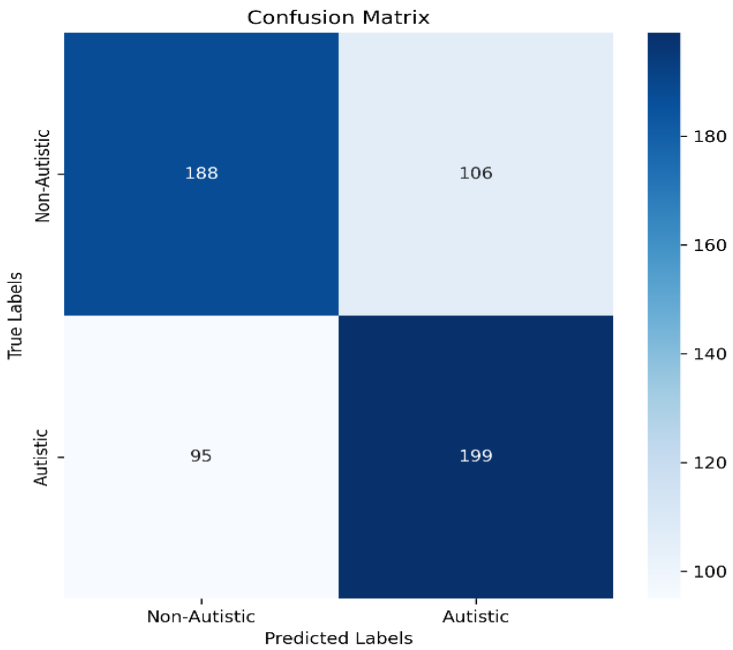
Hyper Parameters Used		
Criterion	**Max Depth**	**Min Sample Split**
gini	10	8

The final model was trained using these parameters, achieving an optimal balance between complexity and performance.



3.2.4 Performance Assessment:

The objective of the evaluation was to assess the Decision Tree Classifier's ability to correctly classify images as autistic or non-autistic using the test dataset. The evaluation was based on several metrics: accuracy, Precision, Recall and Decision Tree which together provided a comprehensive view of the model's performance.



Here is the table summarizing the performance metrics:

Metric	Value
Accuracy	62.07 %
Precision	62.12 %
Recall	61.90 %
F1 Score	62.01 %

These metrics indicate that the model's performance is moderate, with balanced precision and recall, leading to a comparable F1 score.

This breakdown highlights that the model has a balanced performance between the two classes but exhibits notable errors in both categories, indicating room for improvement.

3.2.5 Summary

The model achieved moderate performance, successfully classifying most samples but struggling with false positives and negatives. This outcome suggests that while HOG features and decision trees capture general patterns in the data, they may not be sufficiently nuanced for this complex task. Enhancements, such as using deeper models (e.g., Random Forests or Neural Networks) or incorporating additional features (e.g., color, texture, or contextual information), could improve classification performance.

3.3 Random Forest

3.3.1 Definition

A **Random Forest Classifier** is an ensemble learning method that combines multiple decision trees to improve classification performance. It operates by building a collection of decision trees during the training process and outputs the class that is the majority vote among the individual trees. Each tree in the forest is trained on a random subset of the data, and at each split, a random subset of features is considered, which ensures that the trees are diverse and reduces overfitting.

The key advantage of a Random Forest over a single decision tree is its ability to generalize better by reducing the variance through averaging multiple decision trees. This technique combines the benefits of individual decision trees while mitigating their weaknesses, such as sensitivity to data noise or overfitting. Random Forest classifiers are highly effective in handling both large datasets and complex data structures, and they perform well even with missing data. Furthermore, they are capable of capturing non-linear relationships and feature interactions, making them versatile tools for classification tasks.

Random Forest classifiers are widely used in various applications, from medical diagnoses to image recognition, due to their high accuracy, interpretability, and ability to handle diverse types of data. They are also robust against overfitting, especially when a large number of trees are used, which allows them to perform well on both training and unseen test data.

3.3.2 Feature Extraction

Feature extraction in this project involves converting RGB images into grayscale and resizing them to a fixed dimension of $128 \times 128\ pixels$. Each image is then flattened into a one-dimensional vector, enabling it to serve as input for the Random Forest classifier. This method captures essential pixel-level information and simplifies the representation, making it computationally efficient.

3.3.3 Model Training

The training process was designed to optimize the performance of a Random Forest classifier while maintaining robustness and interpretability.

- **Model Initialization**

The Random Forest classifier was chosen because it builds an ensemble of decision trees, using bagging and feature randomness to improve prediction accuracy and control overfitting. Each tree in the forest makes a prediction, and the majority vote is taken as the final output.

- ***Hyperparameter Tuning***

To achieve optimal performance, the following hyperparameters were tuned using **GridSearchCV**:

- **Number of Estimators (n_estimators)**: The number of trees in the forest, tested at values of 50 and 100.
- **Max Depth (max_depth)**: Maximum depth of each tree, with values of 10 and unlimited depth.
- **Minimum Samples per Split (min_samples_split)**: Minimum samples required to split a node, tested at 2 and 5.
- **Minimum Samples per Leaf (min_samples_leaf)**: Minimum samples required in each leaf node, tested at 1 and 2.
- **Bootstrap**: Whether to use bootstrap sampling, tested as True and False.

GridSearchCV was set up with 3-fold cross-validation to evaluate parameter combinations, ensuring a balance between model complexity and generalization.

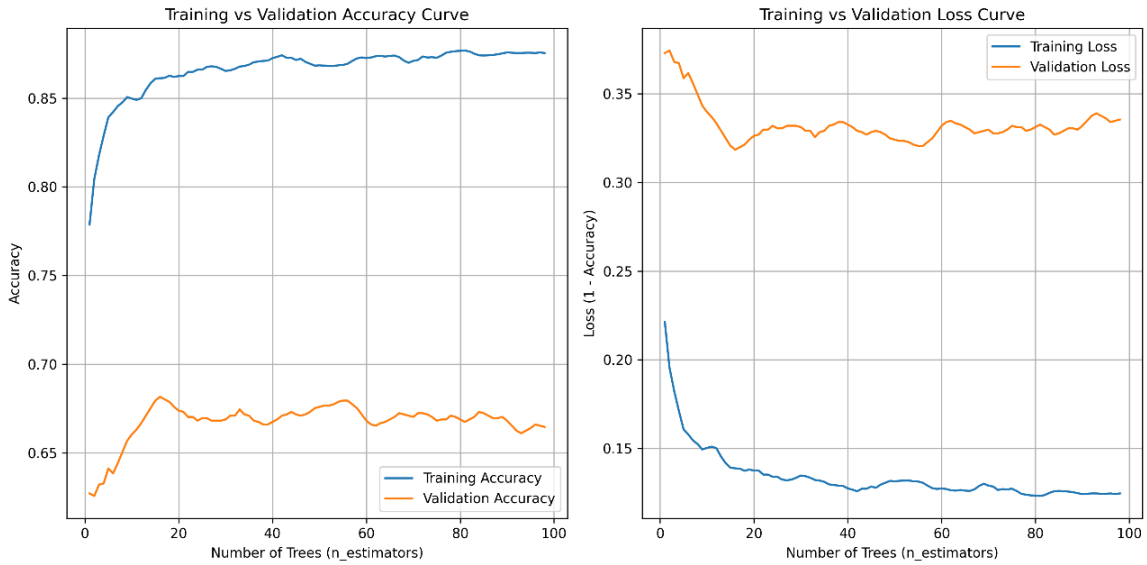
- ***Training with Optimal Parameters***

After tuning, the best parameters identified were:

- n_estimators: 100
- max_depth: None
- min_samples_split: 5
- min_samples_leaf: 1
- bootstrap: True

Hyper Parameters Used				
N_estimators	**Max_depth**	**Min_samples_split**	**Min_samples-leaf**	**bootstrap**
100	none	5	1	TRUE

The final model was trained with these parameters on the training dataset, achieving a robust ensemble that captures intricate patterns in the data.



3.3.4 Model Evaluation

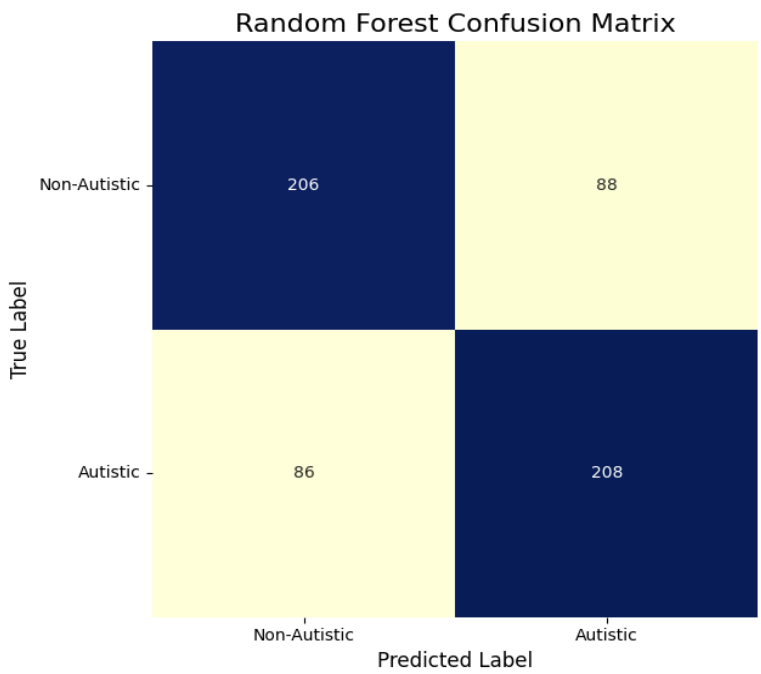
The model's performance was evaluated on the test set using standard classification metrics.

1. Accuracy

The accuracy score of the Random Forest model was **0.6815**, indicating that the model correctly classified approximately **68.15%** of the test samples. This score reflects moderate performance.

2. Confusion Matrix

The confusion matrix revealed the distribution of true positives, true negatives, false positives, and false negatives:



- True positives (autistic images correctly classified): **208**
- True negatives (non-autistic images correctly classified): **206**
- False positives (non-autistic misclassified as autistic): **88**
- False negatives (autistic misclassified as non-autistic): **86**

3. Classification Report

The classification report provided further insights:

- **Precision**: **0.703** (indicates the model's ability to avoid false positives)
- **Recall**: **0.708** (indicates the model's ability to correctly identify all autistic samples)
- **F1 Score**: 0.704 (harmonic mean of precision and recall, balancing false positives and negatives)

These metrics suggest the model achieved a balanced performance between precision and recall, with slight difficulty in handling borderline cases.

Metric	Value
Accuracy	68.2 %
Precision	70.3 %
Recall	70.8 %
F1 Score	70.5 %

3.3.5 Summary

The Random Forest model achieved moderate performance, accurately classifying the majority of samples but encountering challenges with false positives and negatives. This outcome indicates that while the preprocessing steps and Random Forest algorithm effectively captured general patterns in the data, they may lack the sophistication required to discern the finer details of this complex task. Future improvements could include leveraging advanced models like Convolutional Neural Networks (CNNs) to better capture spatial relationships in images or enriching the feature set with additional attributes such as color histograms, texture details, or contextual information. These enhancements could significantly boost the model's classification accuracy and overall reliability.

CHAPTER 4

EARLY DETECTION OF AUTISM USING DEEP LEARNING

4. Deep Learning Model

CNN WITH SELF ATTENTION

The primary goal of this model is to classify grayscale images into one of two binary categories by combining the strengths of convolutional neural networks (CNNs) and self-attention mechanisms. This approach aims to leverage the complementary capabilities of these two components: the local feature extraction prowess of convolutional layers and the global feature understanding enabled by self-attention.

Role of Convolutional Layers

Convolutional layers are well-known for their ability to capture local spatial patterns and hierarchies within an image. They excel at identifying small, localized features such as edges, textures, or specific patterns. In the context of grayscale image classification, these layers help detect fine-grained details or minute anomalies present in the image. This local focus makes CNNs particularly suitable for tasks where detailed analysis of small regions is crucial, such as identifying subtle irregularities in medical images.

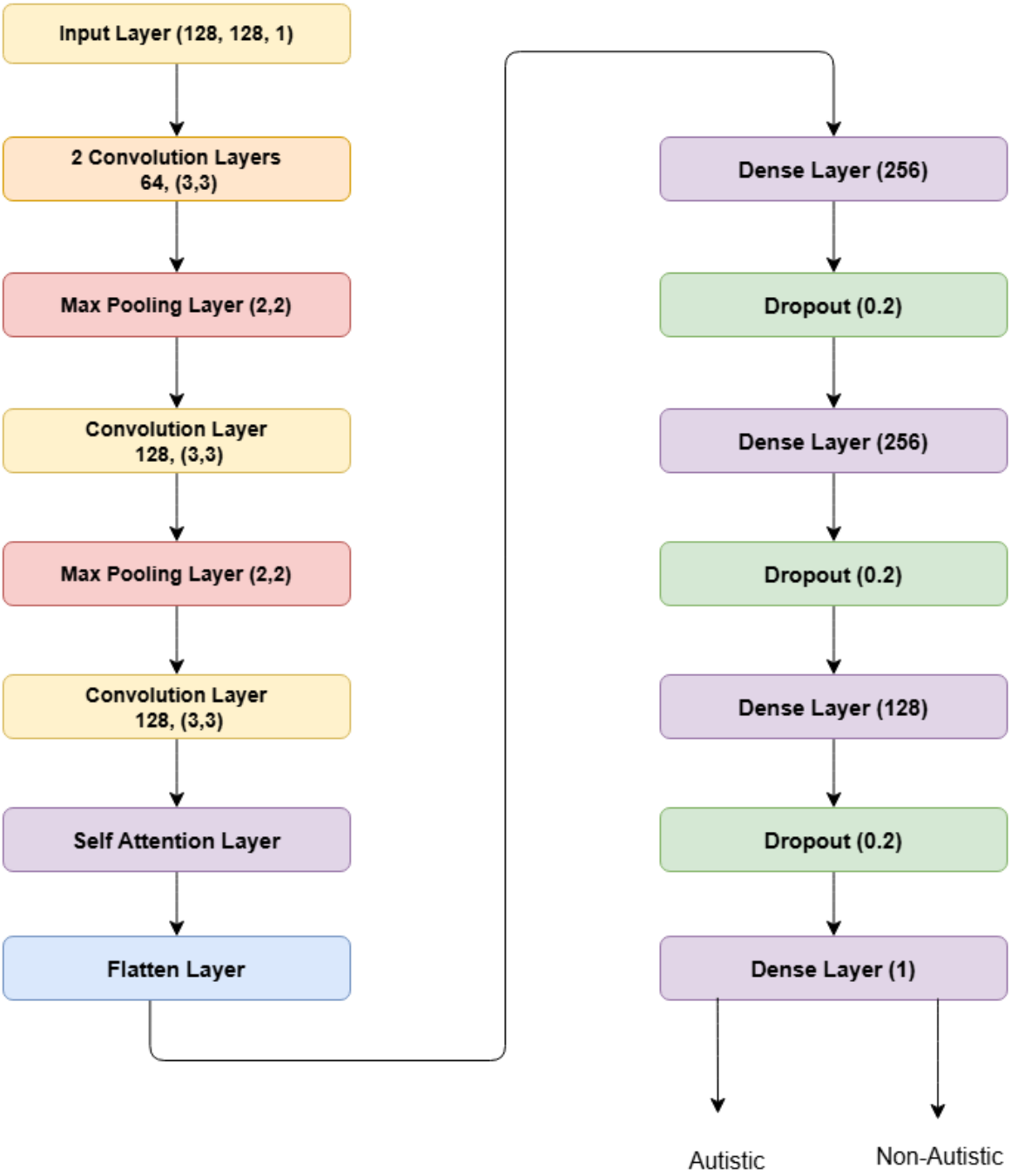
Role of Self-Attention Mechanism

The self-attention mechanism complements the local focus of CNNs by enabling the model to capture long-range spatial dependencies across the entire image. It does so by dynamically weighing the relationships between all spatial locations in the feature map. By doing this, the self-attention mechanism allows the model to focus on important regions and understand how different parts of the image interact, regardless of their distance from each other. This global perspective is essential for tasks where the overall spatial context or relationships between regions significantly influence classification decisions.

Synergistic Integration

By combining these two components, the model gains the ability to both identify detailed features and place them in a broader spatial context. For instance, in medical imaging, the convolutional layers might detect small anomalies or patterns indicative of disease, while the self-attention mechanism can analyze how these anomalies relate to other regions of the image. This integrated approach improves the model's ability to make accurate classifications by simultaneously considering both local and global aspects of the image.

ARCHITECTURE OVERVIEW



LAYERS OF THE ARCHITECTURE

1. *Convolution Layers:*

The first convolutional layer that applies **64** filters of size **3x3** to the input image. This layer is responsible for detecting low-level features such as edges, textures, and simple shapes. The **ReLU** (Rectified Linear Unit) activation function is used to introduce non-linearity, allowing the network to learn more complex patterns. The input shape for this layer is specified as (img_height, img_width, 1), where img_height and img_width correspond to the dimensions of the input images i.e **128**, and 1 represents the grayscale channel.

2. *Max Pooling Layers:*

Max pooling layers are introduced after convolutional layer to reduce the spatial dimensions of the feature maps while retaining the most significant features. Each max pooling operation uses a pool size of **2x2** and a stride of **2**, effectively halving the spatial dimensions. This downsampling reduces computational complexity and enhances the model's ability to generalize by making the network invariant to small translations of the input images.

3. *Self-Attention Layer:*

The **self-attention layer** in this architecture is designed to capture global dependencies within the feature maps, which are output by the preceding convolutional layers. It provides the ability to focus on the most relevant parts of the input feature maps for each output element, improving the model's ability to understand long-range relationships in spatial data.

Unlike convolutions, which focus on small local neighbourhoods, self-attention allows the model to weigh the importance of every spatial location relative to others, enabling it to focus on critical areas.

4. *Flatten Layer:*

The flatten layer converts the 3D feature maps obtained from the convolutional layers into an ID vector. This transformation is necessary to prepare the data for input into the fully connected dense layers, which require ID input. The flattening operation essentially unrolls the spatial dimensions of the feature maps into a single long vector, retaining the information extracted by the convolutional layers.

5. *Dropout Layers:*

Dropout layers are included after each max pooling operation to prevent overfitting. These layers randomly deactivate a fraction of neurons during each training iteration, forcing the model to learn more robust features. A dropout rate of 0.2 is applied after the fully connected layers. Dropout regularizes the network by ensuring that neurons do not become overly reliant on specific other neurons, thus improving the network's generalization capabilities.

6. *Fully Connected Layers:*

A. *Dense Layers*

The first two fully connected layers consist of 256 neurons last fully connected layer contain 128 neurons and uses the ReLU activation function. Those layers acts as a high-level feature aggregator, combining the features extracted by the convolutional and self-attention layer into a meaningful representation that can be used for classification. The ReLU activation function helps in learning complex patterns by introducing non-linearity.

B. *Output Layer:*

The final dense layer uses a sigmoid activation function to produce a probability indicating whether the input corresponds to the "autistic" or "non-autistic" class. Since this is a binary classification task, the sigmoid function outputs a single value between 0 and 1, representing the likelihood of the input belonging to the "autistic" class.

Detailed Explanation of Self-Attention Layer

Working Mechanism

1. Input Feature Map

The feature map from the last convolution layer is given as input to the self-attention layer and has dimensions $(batch_size, height, width, filters)$.

Example: If the batch size is 2, and the feature map dimensions are $8 \times 8 \times 128$, the tensor has shape $(2,8,8,128)$.

2. Query, Key, and Value Generation

The input feature map is passed through three 1×1 convolutional layers $(query_convolution, key_convolution, and\ value_\ convolution)$ to generate:

- **Query**: A transformed version of the input to capture "what to focus on."
$$\boldsymbol{Q} = \boldsymbol{W_Q} * \boldsymbol{X}$$
- **Key**: Another transformed version of the input, representing "reference points for attention."
$$\boldsymbol{K} = \boldsymbol{W_K} * \boldsymbol{X}$$
- **Value**: Encodes the actual information to aggregate based on the attention mechanism.
$$\boldsymbol{V} = \boldsymbol{W_V} * \boldsymbol{X}$$

Where $\boldsymbol{w_Q}$, $\boldsymbol{w_K}$, $\boldsymbol{w_V}$ are filters, and $\boldsymbol{X}$ is the input feature map.

Each operation produces tensors of shape $(batch_size, height, width, channels)$.

3. Attention Scores

- Compute attention scores by taking the dot product of the query and key tensors:
$$\boldsymbol{Attention\ Scores} = \boldsymbol{Query} * \boldsymbol{Key^T}$$
- This results in a tensor of shape $(batch_size, height * width, height * width)$, representing relationships between all spatial positions.
- The scores are scaled by dividing by the square root of the dimensionality of the key vectors ($\sqrt{d_k}$). The division by ($\sqrt{d_k}$) stabilizes gradients.

$$\boldsymbol{Scaled\ Scores} = (\boldsymbol{Attention\ Scores})/\sqrt{(\boldsymbol{d_k})}$$

4. Softmax Normalization

Apply a softmax function to the scores to normalize the scores into attention weights.

$$\boldsymbol{Attention\ Weights} = \boldsymbol{softmax}(\boldsymbol{Scaled\ Scores})$$

5. Weighted Sum of Values

Use the attention weights to compute a weighted sum of the value vectors.

$$\boldsymbol{Attention\ Output} = \boldsymbol{Attention\ Weights} * \boldsymbol{Value}$$

Attention Output aggregates information from all spatial locations for each position, focusing more on regions with higher attention weights.

Advantages of the Architecture

- **Local-Global Feature Extraction:** CNN captures local details, while self-attention captures long-range dependencies.
- **Efficiency:** Use of 1×1 convolutions keep the attention layer computationally efficient.
- **Improved Performance:** Better representation leads to higher classification accuracy, especially for tasks requiring contextual understanding.

Model Training and Evaluation

The training process uses **K-fold Cross-Validation** to train and evaluate this Convolutional Neural Network (CNN) model. The goal of this methodology is to ensure the model generalizes well by validating it on different subsets of the dataset, while also tracking important classification metrics: **accuracy**, **precision**, **recall**, and **F1-score**.

K-Fold Cross-Validation

Definition:

K-fold cross-validation is a technique for assessing the performance of a machine learning model by dividing the dataset into **K** equal-sized folds. The model is trained on **K-1 folds** and tested on the remaining fold. This process is repeated for each fold, ensuring every data point is used for both training and validation. The results are then averaged to provide an overall model evaluation.

- **In this case, K=10**, which means the dataset is split into 10 equal subsets or "folds". Each fold is used as the validation set once, while the remaining 9 folds serve as the training set.
- **Shuffling** is applied before splitting the data to ensure random selection of data points in each fold, which helps reduce bias.

Key Benefits of K-Fold Cross-Validation:

- **Reduced Overfitting:** Each data point is tested against the model during validation, leading to better generalization.
- **More Reliable Metrics:** Since the model is trained and tested on different subsets, the average of the performance metrics (such as accuracy and F1-score) provides a more robust estimate of model performance.

Training and Optimization Steps

After splitting the data into training and validation subsets in each fold, the following training and optimization steps are carried out:

Model Initialization

For each fold, a new instance of the **Convolutional Neural Network (CNN)** is created. This ensures that each fold has a fresh model and avoids the risk of any data leakage between folds.

ModelCheckpoint Callback

A **ModelCheckpoint** callback is used to save the model's weights at the end of each epoch, but only if the model's performance on the **validation set** improves. This is based on the **validation accuracy** metric. The best model, in terms of validation accuracy, is retained, and its weights are saved to a file (specific to each fold).

- **Purpose:** The callback ensures that, after training, only the model with the best validation performance is kept, preventing overfitting to the training data.
- **Validation:** The model is evaluated after each epoch on the validation set, and if the performance improves, the model weights are updated and saved**.**

Hyper Parameters

Parameters	Value
Learning Rate	1e-4
Optimizer	Adam
Batch Size	128
Epochs	60

Accuracy and Loss Curve

Model Evaluation

After each fold's model has been trained, the next step is to evaluate the performance of the trained model on the validation set. The evaluation consists of the following steps:

Predictions

Once the model is trained and its best weights are loaded, predictions are made on the validation set. The predictions are probabilities (between 0 and 1) representing the likelihood of each instance belonging to the positive class. To convert these probabilities into binary classifications (0 or 1), a threshold of 0.5 is applied. Any prediction greater than 0.5 is classified as 1 (positive), and anything less is classified as 0 (negative).

Evaluation Metrics: The performance of the model on the validation set is then evaluated using several key metrics:

1. Accuracy:

- **Definition:** Accuracy is the proportion of correct predictions (both true positives and true negatives) out of all predictions made.
- **Formula:**

$$\boldsymbol{Accuracy = Total\ Predictions\ /\ True\ Positives + True\ Negatives}$$

- **Purpose:** Provides an overall measure of the model's performance.
- **Limitations:** In imbalanced datasets, accuracy can be misleading because the model could achieve high accuracy by simply predicting the majority class.

2. Precision:

- **Definition:** Precision is the proportion of true positive predictions out of all predicted positives.
- **Formula:**

$$\boldsymbol{Precision = True\ Positives\ /\ True\ Positives + False\ Positives}$$

- **Purpose:** Precision measures how many of the predicted positive instances are actually positive.
- **Use Case:** In situations where false positives are more costly (e.g., fraud detection), precision is a critical metric.

3. Recall:

- **Definition:** Recall is the proportion of true positive predictions out of all actual positives.
- **Formula:**

$$\boldsymbol{Recall = True\ Positives/True\ Positives + False\ Negatives}$$

- **Purpose:** Recall measures how many of the actual positive instances were correctly identified by the model.

- **Use Case:** In cases where false negatives are more costly (e.g., disease diagnosis), recall is prioritized.

4. F1-Score:

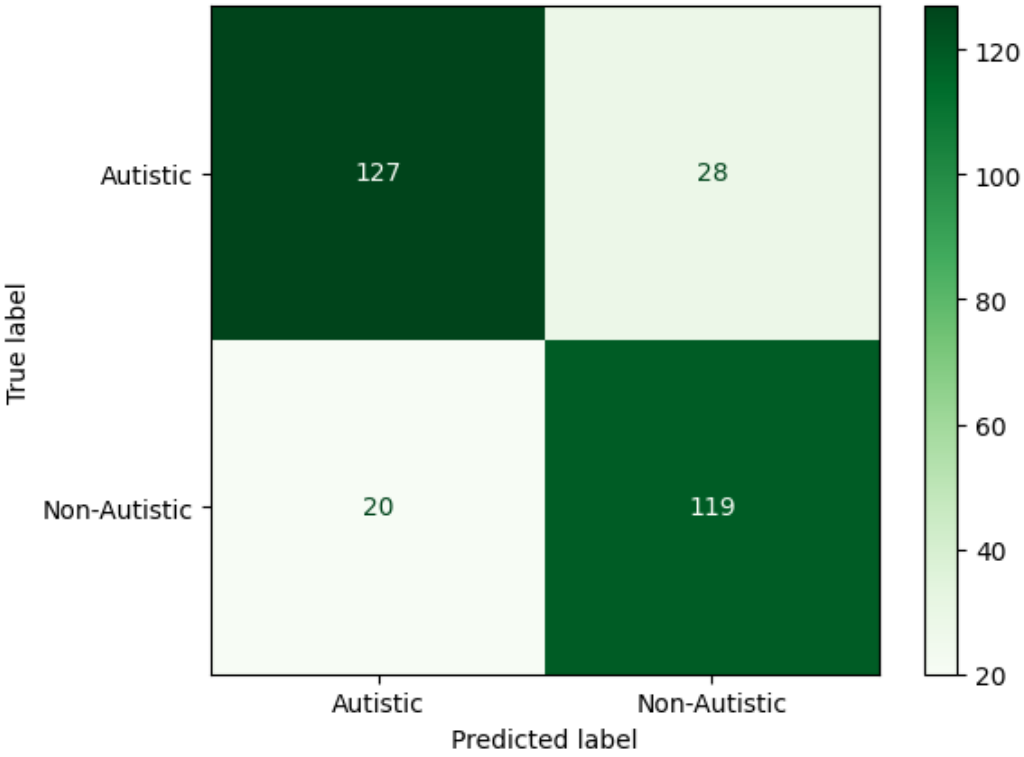
- **Definition:** The F1-score is the harmonic mean of precision and recall. It combines both metrics into a single score that balances the trade-off between precision and recall.
- **Formula:**

$$\boldsymbol{F1 - Score = 2 * Precision * Recall / Precision + Recall}$$

- **Purpose:** The F1-score is particularly useful when there is an imbalance between precision and recall. It provides a single score to evaluate the model's ability to handle both false positives and false negatives.

The metrics (accuracy, precision, recall, F1-score) calculated for each fold are stored in separate lists for later use.

<u>Confusion Matrix</u>



Aggregation of Results

Once all 10 folds have been processed, the final evaluation of the model's performance is computed by taking the **average** of each metric across all folds. This provides a robust estimate of how the model is expected to perform on unseen data.

- **Average Accuracy:** Provides the overall success rate of the model in classifying instances correctly.

- **Average Precision, Recall, F1-Score:** These metrics provide a balanced evaluation of the model's performance in terms of handling both classes, especially when the dataset is imbalanced.

Result over validation set:

Metrics	Value
Accuracy	**83.67 %**
Precision	**81.93 %**
Recall	**86.39 %**
F1 Score	**84.10 %**

Comparison Between the models:

Model	Accuracy	Precision	Recall	F1Score
Logistic Regression	56.14 %	57.0 %	59.0 %	58.0 %
Decision Tree	62.07 %	62.12 %	61.90 %	62.01%
Random Forest	68.2 %	70.3 %	70.8 %	70.5 %
CNN	**83.67 %**	**81.93 %**	**86.39 %**	**84.10 %**

From the above table we can conclude that our custom CNN architecture outperforms other machine learning algorithms.

Chapter 5

Future Scope

5.1 Future Scope

While our study represents a significant step forward in emotion classification for children with ASD, there are several avenues for future research and improvement:

5.1.1 Data Augmentation and Expansion:

Increasing the diversity and size of datasets can help improve model generalization and performance. Additionally, augmenting existing datasets with synthetic data and collecting data from diverse demographic groups can enhance model robustness.

5.1.2 Fine-tuning Pretrained Model Architectures:

Exploring more complex CNN architectures and optimizing hyperparameters could further improve model performance. Techniques such as transfer learning and architecture search can also be leveraged to tailor models to specific emotion recognition tasks.

5.1.3 Enhancing Autism Detection using Custom Architecture:

Future work will involve integrating pretrained models with a novel, custom-designed architecture. By leveraging the strengths of pretrained models for feature extraction and combining them with our purpose-built layers tailored to autism-specific traits, we aim to develop an end-to-end model built from scratch. This approach seeks to significantly enhance accuracy and provide a more robust, generalizable solution for early ASD diagnosis.

5.1.4 Multimodal Emotion Recognition:

Integrating information from multiple modalities such as facial expressions, speech, and physiological signals can provide a more comprehensive understanding of emotions in children with ASD. Future research can explore multimodal fusion techniques to leverage complementary information from different modalities.

5.1.5 Longitudinal Studies and Clinical Validation:

Conducting longitudinal studies and collaborating with clinicians and experts in ASD can provide valuable insights into the practical applicability and effectiveness of emotion recognition systems in real-world settings. Clinical validation and user back are essential steps towards translating research findings into impactful interventions and support tools for children with ASD.

By addressing these challenges and opportunities, future research endeavours can contribute to the development of more effective, accessible, and empathetic technologies for supporting children with ASD in navigating the complexities of social interaction and emotional expression.

Chapter 6

Bibliography

[1] M. F. Rabbi, S. M. M. Hasan, A. I. Champa and M. A. Zaman, "A Convolutional Neural Network Model for Early-Stage Detection of Autism Spectrum Disorder," 2021 International Conference on Information and Communication Technology for Sustainable Development (ICICT4SD), Dhaka, Bangladesh, 2021, pp. 110-114, doi:10.1109/ICICT4SD50815.2021.9397020.

[2] Y. Khosla, P. Ramachandra and N. Chaitra, "Detection of autistic individuals using facial images and deep learning," 2021 IEEE International Conference on Computation System and Information Technology for Sustainable Solutions (CSITSS), Bangalore, India, 2021, pp. 1-5, doi: 10.1109/CSITSS54238.2021.9683205.

[3] Awaji, B., Senan, E. M., Olayah, F., Alshari, E. A., Alsulami, M., Abosaq, H. A., Alqahtani, J., & Janrao, P. (2023). Hybrid Techniques of Facial Feature Image Analysis for Early Detection of Autism Spectrum Disorder Based on Combined CNN Features. *Diagnostics*, *13*(18), 2948. url: https://doi.org/10.3390/diagnostics13182948

[4] T. M. Ghazal, S. Munir, S. Abbas, A. Athar, H. Alrababah, and M. A. Khan, "Early detection of autism in children using transfer learning," *Intelligent Automation & Soft Computing*. doi: 10.32604/iasc.2023.030125.

[5] Alkahtani, H., Aldhyani, T. H. H., & Alzahrani, M. Y. (2023). Deep Learning Algorithms to Identify Autism Spectrum Disorder in Children-Based Facial Landmarks. *Applied Sciences*, *13*(8), 4855. url: https://doi.org/10.3390/app13084855

[6] A. El Mouatasim and M. Ikermane, "Control learning rate for autism facial detection via deep transfer learning," *SIViP*, vol. 17, pp. 3713–3720, Oct. 2023, doi: 10.1007/s11760-023-02598-9.

[7] Mujeeb Rahman, K. K., & Subashini, M. M. (2022). Identification of Autism in Children Using Static Facial Features and Deep Neural Networks. *Brain Sciences*, *12*(1), 94. url: https://doi.org/10.3390/brainsci12010094

[8] Alam, M. S., Rashid, M. M., Roy, R., Faizabadi, A. R., Gupta, K. D., & Ahsan, M. M. (2022). Empirical Study of Autism Spectrum Disorder Diagnosis Using Facial Images by Improved Transfer Learning Approach. *Bioengineering*, *9*(11), 710. url: https://doi.org/10.3390/bioengineering9110710

[9] M. Liao, H. Duan, and G. Wang, "Application of machine learning techniques to detect the children with autism spectrum disorder," *Journal of Healthcare Engineering*, vol. 2022, Article ID 9340027, 10 pages, Mar. 2022. url: https://doi.org/10.1155/2022/9340027.

[10] Akter, T., Ali, M. H., Khan, M. I., Satu, M. S., Uddin, M. J., Alyami, S. A., Ali, S., Azad, A., & Moni, M. A. (2021). Improved Transfer-Learning-Based Facial Recognition Framework to Detect Autistic Children at an Early Stage. *Brain Sciences*, *11*(6), 734. url: https://doi.org/10.3390/brainsci11060734

[11] M. S. Satu, F. Farida Sathi, M. S. Arifen, M. Hanif Ali and M. A. Moni, "Early Detection of Autism by Extracting Features: A Case Study in Bangladesh," 2019 International Conference on Robotics,Electrical and Signal Processing Techniques (ICREST), Dhaka, Bangladesh, 2019, pp. 400-405, doi: 10.1109/ICREST.2019.8644357.

[12] K. S. Omar, P. Mondal, N. S. Khan, M. R. K. Rizvi and M. N. Islam, "A Machine Learning Approach to Predict Autism Spectrum Disorder," 2019 International Conference on Electrical, Computer and Communication Engineering (ECCE), Cox'sBazar, Bangladesh, 2019, pp. 1-6, doi: 10.1109/ECACE.2019.8679454.

[13] Anibal Sólon Heinsfeld, Alexandre Rosa Franco, R. Cameron Craddock, Augusto Buchweitz, Felipe Meneguzzi, “Identification of autism spectrum disorder using deep learning and the ABIDE dataset”, NeuroImage: Clinical, Volume 17, 2018, Pages 16-23, ISSN 2213-1582, url: https://doi.org/10.1016/j.nicl.2017.08.017.

[14] Qin, B., Liang, L., Wu, J., Quan, Q., Wang, Z., & Li, D. (2020). “Automatic Identification of Down Syndrome Using Facial Images with Deep Convolutional Neural Network”. *Diagnostics*, *10*(7), 487. url: https://doi.org/10.3390/diagnostics10070487

[15] S. Jaiswal, M. F. Valstar, A. Gillott and D. Daley, "Automatic Detection of ADHD and ASD from Expressive Behaviour in RGBD Data," 2017 12th IEEE International Conference on Automatic Face & Gesture Recognition (FG 2017), Washington, DC, USA, 2017, pp. 762-769, doi: 10.1109/FG.2017.95.

[16] M. D. Samad, N. Diawara, J. L. Bobzien, J. W. Harrington, M. A. Witherow and K. M. Iftekharuddin, "A Feasibility Study of Autism Behavioral Markers in Spontaneous Facial, Visual, and Hand Movement Response Data," in IEEE Transactions on Neural Systems and Rehabilitation Engineering, vol. 26, no. 2, pp. 353-361, Feb. 2018, doi: 10.1109/TNSRE.2017.2768482.